

Hospital Readmission Risk Predictor

Hyperparameter Tuning & Model Evaluation — Informal Report

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GitHub Repository: <https://github.com/AngelineSetiawan/AngelineSetiawan-hospital-readmission-risk-predictor>

Background & Question (Recap)

Research question: among adult inpatient diabetes encounters, can a supervised classification model identify which patients are at elevated risk of an unplanned readmission within 30 days of discharge, using information available around the time of admission (demographics, prior utilization history, diagnosis groupings, and treatment intensity)?

Hypothesis / prediction: patients with a heavier history of recent healthcare utilization (more prior inpatient stays, more emergency visits) and higher clinical severity indicators (more diagnoses, more procedures, abnormal A1C results) are more likely to be readmitted within 30 days than lower-utilization, lower-severity patients. This is carried over unchanged from the baseline report; nothing in this phase's tuning work motivated a revision to it.

Methods

Recap of Previous Plan

The baseline (Phase 3) model was a logistic regression evaluated at the default 0.50 classification threshold. Its central finding was that 0.50 is unusable given the class imbalance in this dataset: at that threshold the baseline flagged almost no one as high risk (test recall of 0.25%, despite a reported precision of 0.30 on the handful of positive predictions it did make). This phase's plan was to (1) tune the logistic regression itself rather than leave it at default settings, (2) try model families that can capture non-linear interactions between features, and (3) replace the fixed 0.50 threshold with one chosen deliberately from the training data.

Additional Model(s)

Two additional model families were added alongside a properly tuned logistic regression: random forest and XGBoost. Both are tree-based ensembles, chosen because they can capture non-linear and interaction effects among the engineered features that a linear model cannot, and both expose a rich, tunable hyperparameter set (tree depth, leaf size, learning rate, subsampling, class weighting) that lets overfitting and class imbalance be controlled directly rather than assumed away.

All three families were searched with RandomizedSearchCV using PR-AUC (average precision) as the scoring metric, consistent with the baseline report's reasoning that ROC-AUC can look acceptable even when minority-class detection is poor. Searches used stratified 5-fold cross-validation on the training partition only; the held-out test set was not touched until final evaluation. Class weighting (class_weight for logistic regression and random forest, scale_pos_weight for XGBoost) was included as a searchable hyperparameter rather than fixed in advance, so the search itself could decide whether weighting was worth it.

Hyperparameter Tuning

Overfitting control was carried over from the baseline and extended this phase: all hyperparameter selection and threshold selection happen on the training partition, via cross-validation or out-of-fold predictions, never on the test set, so the test set stays a genuinely held-out final check. Hyperparameters tuned per family: logistic regression, penalty (L1/L2), regularization strength C, class_weight; random forest, n_estimators, max_depth, min_samples_leaf, max_features, class_weight; XGBoost, n_estimators, max_depth, learning_rate, subsample, colsample_bytree, min_child_weight, scale_pos_weight.

Threshold Tuning

The classification threshold was tuned separately from the model itself, using out-of-fold predicted probabilities generated by 5-fold cross-validated prediction on the training partition. Both the F1-maximizing and F2-maximizing thresholds were computed. F2 was selected as the operating threshold because it weights recall more heavily than precision, which fits this clinical screening use case: missing a true readmission is more costly than an extra follow-up call to a patient who was not actually going to be readmitted.

Feature Importance Methodology

Permutation importance (the drop in held-out PR-AUC when a single feature is randomly shuffled, averaged over 10 repeats) was used instead of raw model coefficients, so importance is measured on a consistent basis regardless of which family the search selected. This also follows up on a caution the baseline report raised: that raw coefficients for rare medical_specialty categories may reflect small-sample noise rather than a robust clinical association.

Unsupervised Analyses

K-means clustering was fit on the training partition only, using a reduced, clinically interpretable feature subset (utilization counts, treatment intensity, and diagnosis-count fields) rather than the full one-hot-encoded modeling matrix, so the resulting clusters could be described in plain terms to a care-management audience. The number of clusters was chosen from an elbow plot of inertia together with a silhouette-score sweep across $k = 2$ through 8.

PCA was fit on the full one-hot-encoded modeling matrix (training partition only) to check how concentrated the variance is across the engineered feature set and whether a small number of components meaningfully separate patients by readmission status.

Results

Model Family Comparison

Cross-validated PR-AUC by model family, best hyperparameter combination found for each:

Model Family	Best CV PR-AUC	Notes
Logistic Regression (tuned)	0.1603	Selected, narrowly ahead of XGBoost
XGBoost (tuned)	0.1598	Essentially tied with logistic regression
Random Forest (tuned)	0.1542	Lowest of the three

Tuned logistic regression was selected for the remainder of this phase, with an L1 penalty, balanced class weighting, and $C \approx 0.087$, a substantially stronger regularization than the baseline's untuned default of $C = 1.0$. It preserves the interpretability advantage already established in the baseline report and is the model that will be registered for the FastAPI-served pipeline.

Caveat: logistic regression's edge over tuned XGBoost is 0.0005 CV PR-AUC, not a meaningful difference at this search size. The honest read of this result is that model family choice did not matter much this phase; cross-validated PR-AUC also barely moved versus the baseline's own 0.1597, so the search's main value was fixing the threshold rather than raising raw discriminative power.

Threshold Selection

Threshold sweep on out-of-fold training predictions:

Criterion	Threshold	Precision	Recall	F1	F2
Max F1	0.51	0.1439	0.5319	0.2265	0.3455
Max F2 (selected)	0.43	0.1205	0.7861	0.2089	0.3734

Test-set performance (17,610 held-out patients), baseline model versus tuned logistic regression at two thresholds:

Model / Threshold	PR-AUC	ROC-AUC	Precision	Recall	F1	Accuracy
Baseline, 0.50	0.1083	0.6046	0.3000	0.0025	0.0050	0.9316
Tuned, 0.50 (default)	0.1086	0.6080	0.0907	0.5412	0.1554	0.5989
Tuned, 0.43 (selected)	0.1086	0.6080	0.0804	0.7868	0.1459	0.3715

Ranking quality barely moved between the baseline and the tuned model (PR-AUC 0.1086 vs. 0.1083; ROC-AUC 0.6080 vs. 0.6046), so the model's raw ability to separate readmitted from non-readmitted patients is essentially unchanged. What changed is the usable output, and it changed in two separate steps. The first step was the hyperparameter search itself: simply switching to a properly tuned model with balanced class weighting, still evaluated at the default 0.50 threshold, took recall from 0.25% (the baseline's failure mode) to 54.12%. The second step was lowering the threshold to 0.43, which pushed recall further, from 54.12% to 78.68%. Both steps mattered; the search recovered most of the usability on its own, and threshold selection added a further, smaller gain on top. Precision falls to 8.0% at the selected threshold, meaning about 1 in 13 flagged patients is a true positive, and accuracy drops to 0.3715, which is expected and appropriate here since accuracy is dominated by the 93% majority class and is not a meaningful metric for this problem.

Class	Precision	Recall	F1	Support
Not readmitted <30d	0.96	0.34	0.50	16,409
Readmitted <30d	0.08	0.79	0.15	1,201

Relating this back to the research question: the model does carry real, clinically interpretable signal (dominated by A1C testing and result, and prior inpatient history, detailed next), and a low, deliberately chosen threshold makes that signal usable for a screening purpose, catching most readmissions at the cost of many false alarms. Whether this precision and recall balance is the right one for deployment is a clinical and operational decision, and we recommend the care-management team confirm it before the threshold is finalized. The F1-maximizing threshold (0.51, 53% recall, 14% precision) is a readily available, more balanced alternative.

To be explicit about what this model is and is not: the absolute PR-AUC remains modest (0.1086 on the held-out test set, against a base rate of roughly 9.64%), so this should be read as a broad screening and risk-stratification tool, not as a precise readmission predictor. It is well suited to sorting the patient population into higher- and lower-risk groups for follow-up so that limited care-management attention is directed where it is more likely to be needed, and it is not suited to making a confident individual-level call on whether a specific patient will or will not be readmitted. Deployment decisions and stakeholder expectations should be set accordingly.

Feature Importance

Permutation importance (10 shuffles per feature, PR-AUC scoring), computed on the held-out test set:

Rank	Feature	Mean PR-AUC Drop
1	A1Cresult	0.0229

Rank	Feature	Mean PR-AUC Drop
2	A1Cresult_tested	0.0165
3	number_inpatient	0.0128
4	age	0.0033
5	diag_1_group	0.0016
6	number_emergency	0.0015
7	num_procedures	0.0011

A1Cresult and A1Cresult_tested are now the two most influential features by a wide margin, roughly four times more important than any single utilization-history variable, followed by number_inpatient. This meaningfully revises the baseline report's finding, in which medical-specialty categories dominated the largest coefficients: once importance is measured consistently through permutation on held-out data rather than raw coefficients, whether an A1C test was performed and what it showed carries far more weight than specialty categories did. Utilization variables and age remain meaningfully important, consistent with part of the original hypothesis, but are secondary to A1C testing status.

A clarification on interpretation: A1C testing status and result ranking as the strongest predictor is a predictive signal, not a causal one, and should not be read as "ordering an A1C causes readmission." A1C is a routine, standard-of-care test for diabetic inpatients, ordered so clinicians can monitor glycemic control and adjust insulin doses and medication regimens; its predictive strength is best explained by the fact that it is a proxy for how well-controlled a patient's diabetes is at the time of admission. Read this way, the clinically useful takeaway is that patients with the most uncontrolled diabetes, as reflected in an abnormal A1C result, are potentially at higher readmission risk, not that the test itself, or the act of ordering it, drives readmission.

A related caveat on measurement: A1Cresult and A1Cresult_tested are likely correlated, since a recorded A1C result can only exist for a patient who was also tested. With correlated or redundant predictors, permutation importance can split or distort the apparent importance of each one individually, because shuffling a single feature in isolation creates combinations that would not occur naturally alongside the other, unshuffled, feature. The individual importance values reported for these two fields should therefore be read as a combined signal around A1C testing and control rather than as two fully independent findings. We recommend retraining and re-running permutation importance after removing one of the two fields as a follow-up step, to confirm whether the combined signal holds up under one feature rather than being split across both; this is carried forward as a next step below rather than completed in this phase.

Unsupervised Model 1: K-Means Patient Segmentation

K-means was fit on the training partition only, using clinically interpretable utilization and treatment-intensity features rather than the full modeling matrix. A k-sweep from 2 to 8 was run with the elbow method and a silhouette score:

k	Inertia	Silhouette
2	441,200	0.181
3	406,050	0.176
4 (selected)	374,600	0.128
5	349,800	0.126

k	Inertia	Silhouette
6	330,100	0.125
7	314,900	0.121
8	294,300	0.123

Honesty note: silhouette scores are weak throughout (0.12 to 0.18, well below the ~0.5+ that would indicate clearly separated clusters) and are actually highest at k=2 to 3, not at the selected k=4. This is flagged rather than glossed over: the clusters below should be read as soft, overlapping utilization profiles, not as a well-separated typology.

Cluster	Patients	Mean Prior Inpatient	Mean Prior Emergency	30-Day Readmission Rate
0, long-stay, high-procedure	10,384	0.11	0.05	11.50%
1, moderate utilization	21,127	0.09	0.07	9.79%
2, low utilization	18,458	0.06	0.04	6.90%
3, frequent prior utilizer	2,860	2.07	0.81	19.58%

Cluster 3, the smallest group (about 5.4% of training patients), stands out for prior-utilization history rather than current-stay intensity: its average prior inpatient count (2.07) and prior emergency-visit count (0.81) are far above the other three clusters, and its 30-day readmission rate of 19.58% is roughly double the 9.64% overall training rate. Since permutation importance separately identified number_inpatient as the model's third-ranked predictor, this cluster is a credible, independently derived high-risk group. We recommend sharing this cluster definition with care-management as a targeting input, used alongside rather than instead of individual model risk scores.

Unsupervised Model 2: PCA / Dimensionality Check

PCA was fit on the full one-hot-encoded modeling matrix (243 columns, training partition only) to check how concentrated the variance is across the engineered feature set.

Component	Explained Variance Ratio	Cumulative Variance
PC1	0.1042	0.1042
PC2	0.0813	0.1855
PC3	0.0741	0.2596
PC4	0.0585	0.3182
PC5	0.0485	0.3667

The leading component explains only 10.4% of total variance, and the first 20 of the 243 one-hot-encoded columns together explain only 73.9% of variance, indicating the engineered feature set's information is spread broadly across many features rather than concentrated in a small number of directions. Correlation between the leading components and the readmission target was weak across the board (strongest was PC1, at $r \approx 0.071$), so we are not recommending PCA components be added as engineered features at this time.

Update: the corrected calculation has since been re-run successfully, with no errors, and now confirms that 26 components (out of the 243 one-hot-encoded columns) are needed to reach 80% cumulative variance. This is consistent with the broadly spread variance shown in the table above and replaces the provisional note from the earlier version of this report.

Discussion & Next Steps

- The PCA fix is confirmed: the corrected notebook cell ran cleanly with no errors and verified that 26 components are needed to reach 80% cumulative variance. No further action needed on this item.
- Both the hyperparameter search and threshold selection drove this phase's usability gain, in two separate steps. Cross-validated PR-AUC barely moved (0.1597 to 0.1603), but recall at the default 0.50 threshold jumped from 0.25% to 54.12% once the model was properly tuned with balanced class weighting, and lowering the threshold to 0.43 pushed recall further, to 78.68%.
- Route the precision and recall trade-off to the care-management stakeholder audience for an explicit choice between the F2 threshold (0.43, favors recall) and the F1 threshold (0.51, more balanced) before it is finalized for deployment.
- Document A1Cresult and A1Cresult_tested's outsized importance for review, and confirm this reflects genuine clinical signal, specifically a proxy for glycemic control rather than a causal effect of testing itself, and not a data-collection artifact, since these are also the fields the EDA and preprocessing reports flagged for informative missingness.
- Because A1Cresult and A1Cresult_tested are likely correlated, retrain the tuned model and re-run permutation importance with one of the two fields removed, to check whether the combined A1C signal holds up under a single feature rather than being split across both; this was not completed in this phase and is carried forward as a recommendation for the next iteration.
- Revisit K-means k selection given the honesty note above; either justify k=4 on clinical-interpretability grounds explicitly or re-run with k=2/3 as a robustness check.
- Share the Cluster 3 (frequent prior utilizer) definition with care-management as a targeting input, kept separate from individual model risk scores.
- Register the selected tuned logistic regression model and its 0.43 threshold in the MLflow model registry, and build the FastAPI endpoint around it; K-means and PCA remain exploratory artifacts and are not served in production.